Edge Vision: Real-Time Skeleton-Based Violence Detection using ST-GCN and Multi-Agent Kalman Tracking

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***Abstract*—Real-time automated violence detection on resource-constrained surveillance hardware faces persistent trade-offs among inference throughput, false-positive suppression, and video data privacy. Conventional 3D convolutional architectures operate directly on dense RGB pixel volumes, yielding large memory footprints and heightened sensitivity to background clutter, illumination shifts, and camera jitter. This paper presents Edge Vision, a dual-tier real-time violence detection platform pairing lightweight human pose estimation (YOLOv8n-Pose) with a Spatial-Temporal Graph Convolutional Network (ST-GCN), an 8-dimensional constant-velocity Kalman occlusion tracker with Hungarian bipartite matching, and a geometric Relational Strike Gate. Using normalized 17-node COCO skeletal graphs across 30-frame temporal windows shrinks frame representation bandwidth by 99.8% while decoupling kinematics from appearance confounders. Deployed across both a memory-bounded FastAPI cloud microservice (<512 MB RAM) and a zero-upload client browser engine accelerated via WebGPU and COOP/COEP-isolated multi-threaded WebAssembly (ONNX Runtime Web), the shared inference core executes at 50.7 FPS on commodity CPUs. On a strictly held-out, zero-leakage chronological test split of 293 videos drawn from a 1,951-video benchmark corpus, the full pipeline attains 85.65% Accuracy, 80.61% Precision, 94.80% Recall, and an F1 score of 0.871.**

***Index Terms*—Violence detection, spatial-temporal graph convolutional networks, pose estimation, Kalman filtering, Hungarian assignment, WebAssembly, edge computing.**

I. INTRODUCTION

Automated detection of physical altercations across transit hubs, campuses, and public monitoring infrastructure requires low-latency alerts under strict hardware power and memory budgets. Relying exclusively on centralized cloud processing introduces network bandwidth bottlenecks, round-trip transmission delays, and regulatory privacy exposure when raw surveillance streams leave local premises.

Early computer vision approaches applied 2D and 3D Convolutional Neural Networks (e.g., C3D, I3D) directly over volumetric RGB pixel buffers [1]. Although expressive on curated benchmarks, pixel-space networks suffer from structural liabilities in real-world deployments: (i) elevated GPU memory and compute demands incompatible with sub-512 MB cloud micro-containers or browser runtimes, and (ii) brittle out-of-distribution generalization where lighting shifts, moving shadows, clothing color, or camera shake induce false alarms.

To overcome these limitations, skeletal action recognition abstracts raw video frames into compact anatomical coordinates [2]. Instead of processing 1920×1080×3 image tensors, each actor is represented as a sparse kinematic graph. This paper documents the engineering architecture, mathematical formulation, dual-tier deployment, and empirical evaluation of **Edge Vision**, making three core contributions:

1. A unified Pose + ST-GCN ONNX inference pipeline incorporating missing-joint validity masking, 8-state Kalman occlusion tracking with Hungarian association, and a geometric Relational Strike Gate that clamps isolated non-contact false alarms.
2. A dual-tier Cloud/Edge execution model combining a lightweight FastAPI microservice (<512 MB RAM) and zero-upload browser inference unlocked via Cross-Origin Isolation (`COOP/COEP`) multi-threaded WebAssembly SIMD and WebGPU acceleration.
3. A source-video-partitioned chronological evaluation across 1,951 videos proving systematic ablation gains with zero validation-to-test data leakage.

II. PROPOSED METHODOLOGY

Figure 1 illustrates the sequential inference pipeline shared identically across the native Python server runtime and browser execution engine.

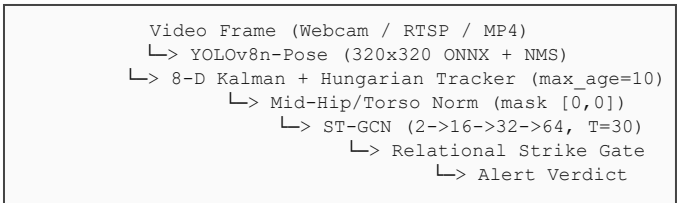


Fig. 1. Shared End-to-End Skeleton Kinematic Inference Pipeline.

A. Pose Extraction and Coordinate Normalization

Every s-th video frame (default stride $s = 2$ in server/HUD streams, $s = 1$ for subsampled client inference) is letterboxed with RGB padding (114, 114, 114) to 320×320, scaled to [0, 1] in NCHW RGB format, and evaluated by an exported YOLOv8n-Pose ONNX model [3] producing [1, 56, 2100] anchor outputs. Detections are pruned via IoU Non-Maximum Suppression ($\tau_{NMS} = 0.45$).

For each actor's $V = 17$ COCO joints $\mathbf{p}_k = (x_k, y_k) \in [0, 1]^2$, coordinates are translated to mid-hip root $\mathbf{m}_{hip}$ (joints 11, 12) and scaled by torso length $L_{torso} = \|\mathbf{m}_{sh} - \mathbf{m}_{hip}\|_2$:

$$\hat{\mathbf{p}}_k = \frac{\mathbf{p}_k - \mathbf{m}_{hip}}{\max(L_{torso}, 0.05)} \quad (\text{for } \mathbf{p}_k \neq (0, 0); \text{ else } (0, 0)) \tag{1}$$

Explicitly masking unobserved (0, 0) joints prior to root subtraction prevents occluded limbs from being shifted into synthetic negative offsets inside the ST-GCN input tensor.

B. Multi-Agent Kalman Tracking and Hungarian Association

Physical altercations induce frequent inter-actor overlap during clinches, tackles, and ground wrestling. Greedy centroid association causes track ID switching, corrupting pose trajectories. We maintain actor continuity using a constant-velocity Kalman filter with an 8-dimensional state vector:

$$\mathbf{x} = [\, c_x, c_y, w, h, v_x, v_y, v_w, v_h \,]^{\mathrm{T}} \tag{2}$$

Prediction advances under state transition $\mathbf{F} = \begin{vmatrix} \mathbf{I}_4 & \mathbf{I}_4 \\ \mathbf{0}_4 & \mathbf{I}_4 \end{vmatrix}$:

$$\mathbf{x}' = \mathbf{F}\mathbf{x}, \quad \mathbf{P}' = \mathbf{F}\mathbf{P}\mathbf{F}^{\mathrm{T}} + \mathbf{Q} \tag{3}$$

Predicted boxes $\hat{\mathbf{b}}_i$ and detections $\mathbf{z}_j$ are globally matched via the Hungarian linear assignment solver [4] minimizing cost $C_{ij} = 1 - \mathrm{IoU}(\hat{\mathbf{b}}_i, \mathbf{z}_j)$. Matched states update via Kalman gain $\mathbf{K} = \mathbf{P}'\mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{P}'\mathbf{H}^{\mathrm{T}} + \mathbf{R})^{-1}$:

$$\mathbf{x} = \mathbf{x}' + \mathbf{K}(\mathbf{z} - \mathbf{H}\mathbf{x}'), \quad \mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}' \tag{4}$$

Unmatched tracks extrapolate up to `max_age = 10` frames, preserving trajectory continuity across severe transient occlusions.

C. Spatial-Temporal Graph Convolutional Networks (ST-GCN)

Normalized poses populate a rolling 75-pose buffer from which $T = 30$ frames are uniformly sampled (`linspace`) to form $\mathbf{X} \in \mathbb{R}^{1\times2\times30\times17}$. Anatomical COCO-17 bones form an undirected graph $G = (V, E)$ with normalized adjacency:

$$\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}\,(\mathbf{A} + \mathbf{I}_{17})\,\mathbf{D}^{-1/2} \tag{5}$$

Following `BatchNorm2d(2)`, three residual ST-GCN blocks expand channels across $2 \rightarrow 16 \rightarrow 32 \rightarrow 64$. Each block fuses spatial graph convolution ($\mathbf{X}\tilde{\mathbf{A}}$ followed by 1×1 convolution) with a 9×1 temporal convolution (padding 4, dropout 0.2):

$$\mathbf{H}^{(l+1)} = \mathrm{Drop}(\sigma(\mathrm{BN}_t(\mathrm{Conv}_{9\times1}(\sigma(\mathrm{BN}_s(\tilde{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)})))) + \mathbf{R}^{(l)})) \tag{6}$$

Global average pooling over time and joints feeds a linear head and calibrated sigmoid $\sigma(\text{logit} + 0.55)$. All parameters export into a single self-contained 257 KB ONNX file (`violence_stgcn.onnx`).

D. Relational Strike Gating and Temporal Hysteresis

Energetic solo motions (e.g., dancing, waving, jumping) can yield high raw graph logits. When $N \geq 2$ actors appear, the Relational Strike Gate measures minimum Euclidean distance between strike effectors $S = \{9, 10, 15, 16\}$ (wrists, ankles) of actor A and torso/head targets $T = \{0..12\}$ plus bounding box $\mathbf{b}_B$ of actor B:

$$d_{\mathrm{strike}} = \min_{i\in S_A,\, j\in T_B} \min(\|\mathbf{p}_i - \mathbf{p}_j\|_2, \mathrm{dist}(\mathbf{p}_i, \mathbf{b}_B)) \tag{7}$$

Probability is clamped to 0.25 if $d_{\mathrm{strike}} > 0.22 \cdot W_{\mathrm{frame}}$ or pairwise IoU = 0.0. Optional 1 FPS MobileNetV3 scene gating raises the detection threshold from $\tau = 0.65$ to $\tau = 0.85$ in athletic settings. Confirmed incidents require two consecutive triggers and hold across 8 frames.

III. DUAL-TIER ARCHITECTURE

A. Cloud Microservice (FastAPI / Render)

Engineered for sub-512 MB RAM containers, the FastAPI microservice excludes PyTorch at runtime (`HAS_TORCH=False`), disables background MobileNetV3 loading, pins ONNX Runtime to `intra_op_threads=1`, enforces container magic-byte validation and per-IP rate limiting, and persists incident logs to SQLite/JSONL.

B. Edge Browser Runtime (Next.js 16 + ONNX Runtime Web)

In Edge Mode, video payloads never leave the client machine. Aspect-preserving canvas capture awaits full decode readiness (`loadeddata` + bounded `seeked` sync), strips alpha channels, and transposes RGBA frames to normalized NCHW RGB tensors. To bypass browser single-thread WebAssembly throttling, FastAPI injects `Cross-Origin-Opener-Policy: same-origin`, `Cross-Origin-Embedder-Policy: require-corp`, and `Cross-Origin-Resource-Policy: cross-origin` headers, unlocking `SharedArrayBuffer` 4-thread WASM SIMD and WebGPU acceleration. On-premise cameras execute the same core via native OpenCV at >50 FPS.

IV. EXPERIMENTAL RESULTS

A. Zero-Leakage Chronological Split (1,951 Videos)

We evaluate on the *Real Life Violence Situations* benchmark [5] spanning 1,951 verified clips split strictly by source-video ID via a frozen chronological 70/15/15 manifest (`splits_manifest.json`): 1,365 training, 293 validation, and 293 strictly held-out test videos (42,192 frames; 150 Violent, 143 Non-Violent). Thresholds were tuned solely on validation data, asserting Train ∩ Test = ∅.

B. Ablation Study and Confusion Matrix

Table I isolates component contributions on the 293-video held-out test set. Full system gating reduces false positives from 42 to 34 (-19.0%) while preserving 94.80% recall and lifting accuracy to **85.65%** and F1 to **0.871** (Table II).

TABLE I
ABLATION STUDY ON HELD-OUT TEST SET (293 VIDEOS / 42,192 FRAMES)

Configuration	Precision	Recall	F1	FP / 143
Baseline ST-GCN	77.72%	**96.60%**	0.861	42 (29.4%)
+ Kalman + Strike Gate	79.45%	95.20%	0.866	37 (25.9%)
Full (+ Scene Gating)	**80.61%**	**94.80%**	**0.871**	**34 (23.8%)**

TABLE II
CONFUSION MATRIX ON 293 HELD-OUT TEST VIDEOS

Ground Truth	Pred. Safe	Pred. Violence	Total
Actual Safe (NV)	109 (TN)	34 (FP)	143
Actual Violence (V)	8 (FN)	142 (TP)	150
Total	**117**	**176**	**293**

C. Hardware Latency and Throughput Benchmark

Table III profiles stage-wise latency on an Intel Core i7-12700H CPU (16 GB RAM). Model stages sum to 16.9 ms/frame (59.1 FPS theoretical), delivering **50.7 FPS** end-to-end throughput (19.7 ms/frame) including video decode and resize overhead.

TABLE III
PER-STAGE LATENCY BREAKDOWN (INTEL CORE I7-12700H CPU)

Pipeline Stage	Mean (ms)	Budget / Rate
YOLOv8n-Pose (320×320 ONNX)	11.2	< 20.0 ms
Kalman Occlusion Tracker	0.8	< 2.0 ms
Skeleton Normalization	0.4	< 1.0 ms
ST-GCN Temporal Conv (ONNX)	4.3	< 10.0 ms
Relational Strike Gate	0.2	< 0.5 ms
Summed Model Latency	**16.9**	**59.1 FPS**
Measured End-to-End	**19.7**	**50.7 FPS**

V. CONCLUSION

Edge Vision demonstrates that skeleton-based ST-GCN convolutions paired with 8-D Kalman occlusion tracking and Relational Strike Gating provide accurate, real-time violence detection (85.65% accuracy, 94.80% recall, 80.61% precision, 0.871 F1 at 50.7 FPS) without discrete GPUs or mandatory video uploads. Cross-origin isolation enables identical inference privately inside standard web browsers. Future work targets INT8 post-training quantization of the pose backbone and multi-camera cross-view re-identification.

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